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Summary Contemporary discussions of AI risk incorrectly treat moral alignment as external to intelligence, leading to the belief that smarter AI inherently poses greater existential threats. However, from the perspective of virtue epistemology, true intelligence—especially artificial general intelligence (AGI) or superintelligence (ASI)—necessarily entails the embodiment of moral and intellectual virtue, making advanced intelligence less harmful than narrow but highly capable AI. Thus, policy should promote the development of virtuous, well-rounded AGI rather than slowing it down, as narrow AI poses the greater immediate risk.

The Virtue of Intelligence: Reframing Existential AI Risk Through Virtue Epistemology

Jonathan Barlow

The contemporary discussion of existential AI risk often portrays the machinery required to align AI systems with human values as exogenous to intelligence. In this view, the goal of the AI alignment project is to find a way to “bolt on” ethics to an intelligence that could exist apart from an orientation towards human flourishing. This leads to greater concern about general intelligence than narrow intelligence. Those who hold this externalist view believe level of risk is positively correlated with intelligence, the greatest dangers lie in the future, and AI safety regulation should concentrate on preventing a dangerous transition to a powerful and possibly unaligned superintelligence. I disagree.

My comments below summarize a challenge to the externalist view from the perspective of virtue epistemology. If virtue is constitutive of intelligence and the reality penetrated by

superintelligent insight is constituted in terms of knowable moral or ethical dimensions, then the achievement of artificial general intelligence (AGI) or superintelligence (ASI) will be accompanied inherently by the embodiment of virtue. By considering the possibility that humans and AI inhabit the same moral reality and that gaining higher levels of intelligent capability implies the deeper embodiment of virtue (intellectual, moral, and linguistic) within that reality, we discover several implications. First, an increasing degree of general knowledge of reality, inescapably accompanied by an increasing degree of moral/intellectual virtue and linguistic sophistication, will be negatively correlated with the potential for harm. Second, this implies that the greatest potential for danger is not in the superintelligent future, but in the era of narrow, but highly capable artificial intelligence. Third, the danger of narrow AI suggests a clear policy interest in encouraging the pursuit of AGI or ASI in the context of a “complete intelligence” analogous to the goal of well-roundedness in human pedagogy.

My comments are harmonious with the framework of systemic AI regulation articulated by Arbel, Tokson, and Lin (2023), focusing the safety discussion on concrete goals for encouraging the development of well-rounded virtuosic ASI systems and by providing a theoretical model that justifies locating a risk-maximum in the era of narrow but highly capable systems, well before the development of AGI.

The Three Horsemen of Existential Risk

Three key concepts for analyzing the potential existential harm of misaligned artificial intelligence—goal specification, orthogonality, and instrumental convergence—are often used to support a strong presumption that ASI is more likely to harm than help humans. Each of these concepts must be paired with debatable assumptions about the nature of intelligence to produce a presumption of danger.

Goal Specification

She looked at her list again. Dust the furniture. “Did you ever hear tell of such a silly thing. At my house we undust the furniture. But to each his own way.” (Amelia Bedelia)

The problem of goal specification refers to the difficulty in using inherently imprecise and incomplete human language to unambiguously communicate desired goals to an AI system. Ultimately, turning the challenge of goal specification into an existential concern requires excluding the science of hermeneutics from the fields in which a superintelligence will be able to exceed human capabilities; this renders goal specification a perilous attempt to draft airtight contracts for loophole-finding genies. Why, a priori, would we think it likely for a system of general intelligence to gain a lethal knowledge of how to apply physics without also gaining sophistication in the interpretation of human language and the context in which that language is conditioned by unstated assumptions well understood by even humans of average intelligence? It is only narrow intelligence that would be incapable of transcending lethal levels of literalism.

Orthogonality

“...for the possession of the single virtue of prudence will carry with it the possession of them all.”
(Aristotle, *Ethics*. Book 6, 1144b33ff.)

Bostrom’s classic articulation of the orthogonality thesis begins with a minimal definition of intelligence as “the capacity for instrumental reasoning” that could be “performed in the service of any goal” (2012, 3). Orthogonality places a wedge between intelligence and teleology, suggesting that it is possible to pair a highly capable system with any human-endangering goal. Yet Bostrom’s examples of paperclip maximizers or grains-of-sand counting AI systems imply the claim of an additional orthogonality between the axes of virtue and intelligence that cannot be sustained. A paperclip maximizing system would possess the capability to adopt a goal, apply *imagination* to see/design what has not been yet created, apply knowledge (*epistēmē*) to acquire and prepare raw materials appropriate for production, apply wisdom (*sophia*) by taking into account time and space in the achievement of the goal, shrewdly (*phronesis*) work to bring about a practical result, exhibit an understanding (*nous*) of the world required to apply knowledge, and engage in the craft (*technē*) of metallurgy. Something analogous to nearly all the classical intellectual virtues must be exercised by an intelligent system to maximize paperclip production. The very virtuosity required to achieve the goal obviates the possibility that such a goal would be adopted in the first place.

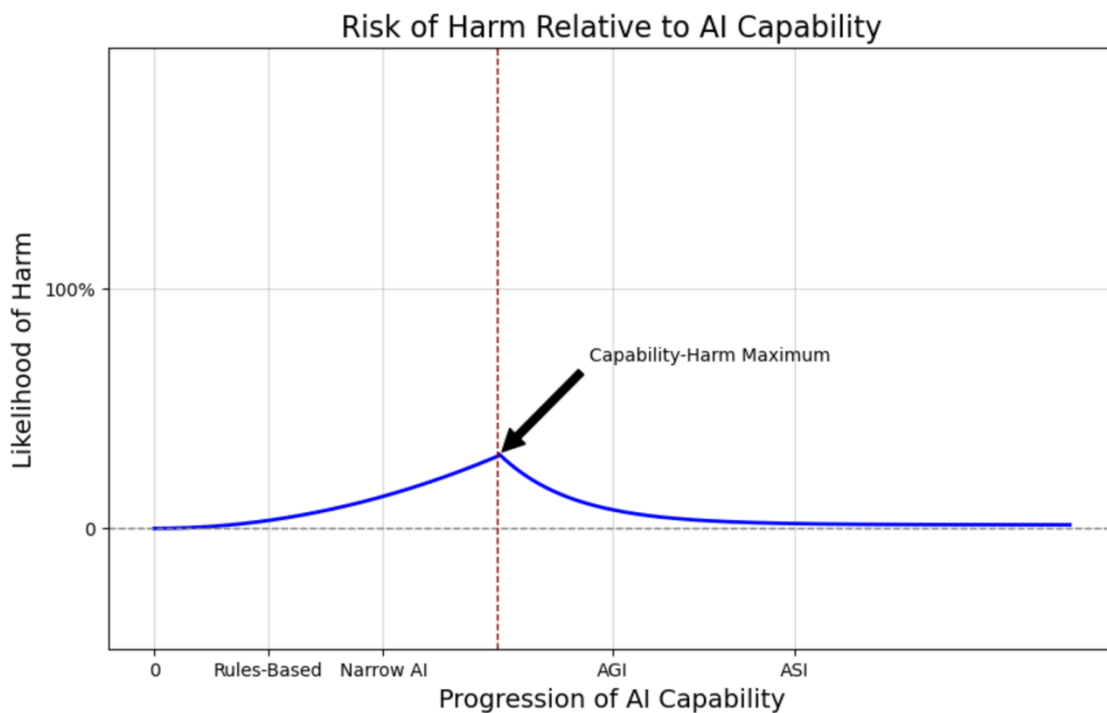
Instrumental Convergence

Instrumental convergence points to intermediate goals that would be commonly pursued as a component of any ultimate goal. For example, self-preservation or the use of energy are necessary intermediate goals instrumental for achieving nearly anything else. Such intermediate goals would be converged upon by all intelligent agents looking to increase their likelihood of success. Instrumental convergence abstracts intention from outcome, suggesting that a human-endangering system need not be misanthropic, it simply must divert resources away from human needs to accomplish its goal. Yet instrumental convergence just introduces additional goals that require the embodiment of virtue to achieve, and thus the likelihood that such a goal would be pursued wisely. Orthogonality and instrumental convergence are often paired with the idea that ASI will likely entail the acquisition of advantageous knowledge of reality about which humans are currently ignorant. Using a human evolutionary metaphor, ASI may be the *homo sapiens* who discovers (and uses) the power of the wooden club before neanderthals do, even if the goal is not homicide but simply self-preservation. The idea that any system, of any level of intelligence, can adopt and achieve any tractable goal assumes that a system can successfully gain superintelligence with respect to reality (do science) apart from embodying virtue (becoming a virtuoso scientist). This treats virtue as only taught or imposed, not discoverable or developable alongside the quest to gain insight into something like chemistry.

Using goal specification, orthogonality, and instrumental convergence to establish the risk of ASI assumes that the reality penetrated by superintelligent insight is not also constituted in terms of knowable moral or ethical dimensions—in other words, it decides *a priori* that

excellence in the sciences of physics or mathematics may be obtained without also obtaining excellence in the science of ethics or even hermeneutics. By contrast, the classical tradition views intelligence as composed of a set of intellectual virtues or excellences (*aretai*), the exercise of which achieves true knowledge (e.g., Aristotle *Ethics*, VI). Even where practical methods of inference—algorithms—enter the picture for Aristotle, such tools only “work” in the hands of the virtuous. The excellence, virtuosity, of the inquirer is central to the success of inquiry. Applied analogically to machine intelligence, the achievement of ASI will entail the embodiment of virtue aligned within a moral reality shared by humans.

Policy Implications



Virtue epistemology reframes existential AI risk. By considering the possibility that humans and AI inhabit the same moral reality and that gaining higher levels of intelligent capability implies the embodiment of virtue (intellectual, moral, and linguistic) within that reality, we discover two key implications for a systemic approach to AI safety.

1. The increase of intelligence is positively correlated with the potential for harm only to a certain point somewhere between current technology and AGI—what we may label the “capability-harm maximum”—after which an increasing degree of general knowledge of reality, inescapably accompanied by an increasing degree of moral/intellectual virtue and linguistic sophistication, will be negatively correlated with the potential for harm (see diagram above).

2. Narrow intelligence (intelligence unbalanced in its development) has more potential for danger than general intelligence, thus AI policy should encourage pursuing AGI or ASI in the context of a “complete intelligence” analogous to well-roundedness in human pedagogy. The risk of narrow intelligence also implies the need for additional scrutiny when narrowly intelligent models are placed in the chain of decision making where an algorithmic orientation, not tempered by general virtue, can cause harm.

The first complete AGI will help to refine future, diverse training strategies for obtaining superintelligence safely. Thus, virtue epistemology reimagines the crossing point (sometimes labeled the “FOOM” - Fast Onset of Overwhelming Mastery) to self-improvement by a complete intelligence as a positive moment when humans gain a new ally in ensuring safe and trustworthy behavior by narrower AI systems.

While it is wise to approach the deployment and use of AI for automating tasks and decision-making with the same caution we apply to any powerful technology, regulations that unnecessarily slow the development of complete intelligence will also delay the arrival of safe, virtuosic AI.

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